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CONTROL MANIFOLD AND METHOD FOR TANK CLEANING

Abstract

An independent control manifold is configured to handle supplying a cleaning fluid to, and extracting a sediment from an operational tank, The control manifold includes a supply block configured to receive the cleaning fluid from an external first tank and to supply the cleaning fluid to the operational tank; a discharge block configured to receive the sediment from the operational tank and to supply the sediment to an external second tank; and a venturi eductor that fluidly connects the supply block to the discharge block, and is configured to either extract the sediment from the operational tank or to recover the cleaning fluid.

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Background/Summary

BACKGROUND OF THE INVENTION

Technical Field

[0001] Embodiments of the subject matter disclosed herein generally relate to an independent control manifold, associated system, and method for removing sediment from inside a tank using the control manifold.

Discussion of the Background

[0002] In the chemical field, for example, oil and gas refineries, there are many operations that require storing in a container (e.g., a tank or reservoir) one or more fluids for certain amounts of time or for processing the one or more fluids, and then providing the one or more fluids to another component of the refinery for further processing. These fluids are sometimes not pure fluids, i.e., they include one or more other particulates, which tend to sediment in time. Thus, in time, the tanks or reservoirs where these fluids are customarily stored or processed, start to accumulate these particulates as residue (called herein “sediment”). Over enough time, the amount of sediment may negatively interfere with the processing operations of the refinery (or any other plant). While this specific example refers to oil and gas processing, most chemical processing industries experience the same problems. In fact, even industries like food processing or pharmaceuticals are facing the same problems.

[0003] To remove the sediment, the assignee of this patent application has developed a specialized slurry dispensing system (also called cleaning device or cleaning system in this document) that can be placed inside the tank for removing the sediment with minimum disruption to the processing of the fluid. This system, called herein the Octopus, is described in International Patent Publication Application no. PCT/US2023/010320, filed on Jan. 6, 2023, the entire content of which is incorporated herein by reference. This patent application discloses that the Octopus system is attached to the interior of the tank by using various mechanical or magnetic means. Other cleaning systems for removing the sediment from a tank are today in use and compatible with the embodiments that are discussed later.

[0004] All the traditional cleaning systems need to be connected to a water source and also to a sediment collection system for removing the sediment from the container. For example, FIG. 1 illustrates a connecting system **100** that provides water and offers storage space for the sediment. The connecting system **100** may be fluidly connected to one or more cleaning devices **160-1**, **160-2** (for example, Octopus, which are generically indicated by reference number **160-I**, where I may be any positive integer equal to or larger than 1), which are located inside an operational tank **162**.

[0005] The connecting system **100** includes a motive water delivery system **110** and a slurry receiving system **130**. The delivery system **110** includes a water tank **112** that holds the water necessary for cleaning the inside the operational tank **162**. The water from the water tank **112** is pumped by a pump **114** through various piping **116** and intake valves **118** to each cleaning device **160-I**. The receiving system **130** includes a receiving tank **132** for storing the sediment that is removed from the operational tank **162**. Plural pipes **134** and discharge valves **136** fluidly connect the discharge of the cleaning device **160-I** to the receiving tank **132**. Thus, each cleaning device **160-I** is fluidly connected to a water delivery pipe **120** of the delivery system **110** and a sediment removing pipe **138** of the connecting system **100**.

[0006] However, the existing connecting systems **100** are expensive (requiring a significant capital investment), static (difficult if not impossible to be moved to another tank), are difficult to be customized, and require local water availability and local sediment storage facilities. Thus, there is

a need for a new connecting system, among the cleaning devices, a water source, and a sediment storage tank, that overcomes the above noted limitations.

SUMMARY OF THE INVENTION

[0007] According to an embodiment, there is an independent control manifold for handling supplying a cleaning fluid to and extracting a sediment from an operational tank. The control manifold includes a supply block configured to receive the cleaning fluid from an external first tank and to supply the cleaning fluid to the operational tank, a discharge block configured to receive the sediment from the operational tank and to supply the sediment to an external second tank, and a venturi eductor that fluidly connects the supply block to the discharge block, and is configured to either extract the sediment from the operational tank or to recover the cleaning fluid.

[0008] According to another embodiment, there is a cleaning truck for supplying a cleaning fluid and extracting a sediment from an operational tank. The cleaning truck includes a tank for storing the cleaning fluid and the sediment and a control manifold located outside the tank. The control manifold includes a supply block configured to receive the cleaning fluid from the tank and to supply the cleaning fluid to the operational tank, a discharge block configured to receive the sediment from the operational tank and to supply the sediment to the tank, and a venturi eductor fluidly connecting the supply block to the discharge block, and configured to either extract the sediment from the operational tank or to recover the cleaning fluid from the tank.

[0009] According to yet another embodiment, there is a method for removing a sediment, with a control manifold, from an operational tank. The method includes supplying a first cleaning fluid, with a supply block of the control manifold, to an inlet of the operational tank, discharging the sediment, with a discharge block of the control manifold, from an outlet of the operational tank, diverting a part of the first cleaning fluid, from the supply block to a first inlet of a venturi eductor to create a vacuum at a second inlet, and pulling in a fluid, at the second inlet of the venturi eductor.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] For a detailed description of the disclosed, representative embodiments, reference will now be made to the accompanying drawings, which include the following figures, wherein:

[0011] FIG. 1 is a schematic diagram of a cleaning system located in a tank and a connecting mechanism that provides water to the cleaning system and stores the sediment removed from the tank by the cleaning system;

[0012] FIG. 2 is a schematic diagram of a system that uses two trucks for cleaning a tank with a cleaning system located inside the tank;

[0013] FIG. 3 is a schematic diagram of another system that uses two trucks and an independent control manifold for connecting the two trucks to a cleaning system located inside the tank;

[0014] FIG. 4 is a schematic diagram of the independent control manifold and its connections to two trucks and the cleaning system according to an embodiment;

[0015] FIG. 5 is a schematic diagram of the independent control manifold and its connections to two trucks and the cleaning system according to another embodiment;

[0016] FIG. 6 is a schematic diagram of the independent control manifold and its connections to two trucks, an external tank, and the cleaning system according to an embodiment;

[0017] FIG. 7 is a schematic diagram of the independent control manifold and its connections to a single truck, an external water supply, and a cleaning system according to an embodiment;

[0018] FIG. 8 is a schematic diagram of the independent control manifold and its connections to a fixed water supply, an external tank for storing a sediment, and a cleaning system according to an embodiment;

[0019] FIG. **9** is a schematic diagram of the cleaning system;
[0020] FIG. **10** is a schematic diagram of the control manifold being attached to a truck according to an embodiment;
[0021] FIG. **11** is a schematic diagram of the control manifold of FIG. **10** being attached to the truck;
[0022] FIG. **12** is a schematic diagram of the control manifold of FIG. **10**;
[0023] FIG. **13** is a schematic diagram of the truck, its control manifold, and the various connections to an operational tank; and
[0024] FIG. **14** is a flow chart of a method for cleaning an operational tank with the control manifold of any of FIGS. **3** to **12**.

DETAILED DESCRIPTION OF THE INVENTION

[0025] The following description of the embodiments refers to the accompanying drawings. The same reference numbers in different drawings identify the same or similar elements. The following detailed description does not limit the invention. Instead, the scope of the invention is defined by the appended claims.

[0026] Reference throughout the specification to “one embodiment” or “an embodiment” means that a particular feature, structure or characteristic described in connection with an embodiment is included in at least one embodiment of the subject matter disclosed. Thus, the appearance of the phrases “in one embodiment” or “in an embodiment” in various places throughout the specification is not necessarily referring to the same embodiment. Further, the particular features, structures or characteristics may be combined in any suitable manner in one or more embodiments.

[0027] In various embodiments, an independent control manifold is provided on a movable platform (called herein simply the “platform”) and this assembly (called “control manifold assembly”) may be dispatched, as needed, to an operational tank for removing the inside sediment. The control manifold assembly may be fluidly connected to a truck (e.g., water supply truck, and/or sediment storage truck) for distributing the water to the cleaning systems located inside the operational tank. The control manifold assembly may handle only the water supply part, only the sediment removal part, or both the water supply and the sediment removal parts. Details of the control manifold assembly and various possible implementations of it are now discussed with regard to the figures.

[0028] FIG. **2** illustrates the control manifold assembly **200** having the independent control manifold **210** supported (for example, with a frame **211**) on the platform **230**. Note that the term “independent” is used herein to indicate that the control manifold assembly **200** or its control manifold **210** are independent of the trucks, i.e., they can be moved where desired without the need of the trucks. The platform **230** may be a skid, for example, having a frame **232** and wheels **234**. The frame **232** may have a connecting mechanism **236** for connecting to a vehicle for being towed. The control manifold **210** may include pipes **212**, a measurement device **214** (for example, a flow meter, pressure device, temperature sensor, etc.), various valves **216**, connection ports **218**, sight element **220**, and choke valve **222**. The connection ports **218** (e.g., female and/or male camlocks) are used to connect to various houses of the trucks or the operational tank, and/or the cleaning systems. The sight element **220** includes at least a transparent wall portion so that the operator can visually inspect the fluid/sediment being pumped through the control manifold. The choke valve **222** is used to control and/or suppress the flow of fluids through the control manifold. More elements may be added to the control manifold **210** as discussed later.

[0029] FIG. **2** shows a water supply truck **260** having a water tank **262**, a pump **264** for pumping the water from the tank, and a flexible pipe (e.g., a hose) **266** for fluidly connecting the pump **264** to the cleaning device **160-I** to supply the water. The figure shows connecting elements **268** and corresponding intake valves **118** connected along the water flow to the functional tank. FIG. **2** further shows a sediment storage truck **270** having a sediment storage tank **272** and a connecting hose/pipe **274** for fluidly connecting to the control manifold **210**. The control manifold **210** is also

fluidly connected, through a discharge conduit **280**, to connecting elements **282** and corresponding discharge valves **136**, which are fluidly connected to the cleaning devices **160-I** inside the functional tank **162**. The connecting elements **268** and **282** and the valves **118** and **136** are fixedly attached to the operational tank **162**. Due to the above discussed connections, the water pumped from the water supply truck **260** reaches the cleaning devices **160-I**, dislodges the sediment from the operational tank **162** and forms a slurry, and the slurry is discharged through the discharge valves **136** to the control manifold assembly **200** and then moved to the sediment storage truck **270**, where it is stored. Note that in one embodiment, the pressure existing in the operational tank **162** makes the slurry to move along the path described above. A bypass valve **284** may directly connect the intake valves **118** to the discharge valves **136** for backflush or other purposes.

[0030] The cleaning system **200** illustrated in FIG. 2, e.g., control manifold assembly **200**, water delivery truck **260**, and sediment storage truck **270**, requires that the trucks leave the job site to either get more water or unload the stored sediment, which is time consuming and expensive. Thus, according to an embodiment, as schematically illustrated in FIG. 3, a shuttle truck **360** is used to travel to a source of water and bring water for the water supply truck **260**. The shuttle truck **360** has its own tank **362** for carrying the water and its own pump **364** (e.g., pressure pump, vacuum pump, or both) for pumping the water. The shuttle truck **360** may also be used to remove the sediment from the sediment storage truck **270** so that the sediment storage truck **270** does not have to leave the operational tank **162**.

[0031] Thus, in one embodiment, the shuttle truck **360** is initially supplied with a full load of water. As the water truck tank **262** of the water supply truck **260** empties, water from the shuttle truck **360** is pumped into the water supply truck **260** to replenish its supply so that it can continue operating. After the shuttle truck **360** pumps its load of water into the water supply truck **260**, it then uses its vacuum pump **364** to pull water from the sediment storage truck **270**, which is receiving the sediment slurry from the cleaning device **160-I**. Note that the slurry discharged from the operational tank **162**, into the tank **272** of the sediment storage truck **270**, separates due to the gravity into the sediment **271**, which accumulates to the bottom of the tank **272**, as schematically illustrated in FIG. 3, and a cleaning fluid (e.g., the clean water) **273**, which stays at the top of the tank **272**. Due to this separation, the shuttle truck **360** can pump this load of water back into the water supply truck **260**, when needed. With this method, three trucks **260**, **270**, and **360** can be utilized to operate multiple cleaning devices **160-I**, until the sediment storing truck **360** contains the maximum weight of solids that it can carry.

[0032] In another embodiment, as illustrated in FIG. 4, the control manifold **210** is configured to handle not only the discharge fluid from the operational tank **162**, but also the water supply. In this embodiment, the control manifold **210** includes a supply block **410**, a discharge block **430**, and a venturi eductor **450** fluidly connected to each of the supply block and discharge block. The venturi eductor is a device having two inlets and an outlet. A first inlet is used to receive a first fluid (e.g., water), under pressure for generating a vacuum at a second inlet. The second inlet may be connected to any source of a second fluid (in one embodiment, the second fluid is the same as the first fluid, e.g., water; however, in another embodiment, the second fluid is the slurry that includes the sediment). The output discharges a mixture of the first and second fluids. Note that while the first inlet acts as the “engine” of the venturi eductor, the second inlet and the outlet act as a “vacuum cleaner.”

[0033] The control manifold **210** optionally includes the support **211**, for being attached to the platform **230**. The supply block **410** includes an inlet port **412** for fluidly connecting to the flexible pipe **266** providing water from the water supply truck **260**. The inlet port **412** may be implemented as a camlock or any other known lock. The inlet port **412** is fluidly connected to a water supply pipe **414** that extends along the control manifold **210**. The water supply pipe **414** may be provided with a control valve **416** that may be used to control the water flow entering the operational tank **162**. The water supply pipe **414** terminates with one or more outlet ports **418-I**, where I may be the

number of cleaning devices **160-I** located inside the operational tank **162**. However, **I** may be larger or smaller than the number of cleaning devices inside the operational tank. The outlet ports **418-I** are configured to fluidly connect to corresponding inlet ports **164-I** of the operational tank **162**. These ports are fluidly connected to inlet ports (not shown) of the cleaning devices. While FIG. 4 shows the presence of four cleaning devices **160-I**, fewer or more cleaning devices may be used. The water supply pipe **414** may have a T-junction **420** that fluidly communicates, through another control valve **422**, with the venturi eductor **450**. By using the control valves **416** and **422**, the operator of the control manifold **210** may control the amount of water flowing toward the cleaning devices **160-I** and toward the eductor **450** (for example, only to the cleaning devices, only to the eductor, or to both of them in a desired split). The purpose of this control is discussed later. The supply block **410** may also include one or more sensors **214**, e.g., pressure sensor, flowmeter, etc. [0034] The discharge block **430** includes the discharge pipe **212**, one or more sensors **214**, a sample valve **216** (for sampling the discharged slurry), an outlet port **218** for connecting to the discharge conduit **280** (e.g., a hose, flexible or solid) and to the tank of the sediment storing truck **270**. The discharge block **430** may also include the sight element **220**, and the choke valve **222**. The discharge pipe **212** is also connected to one or more inlet ports **224-I**, where **I** may describe the number of cleaning devices **160**. However, the number of inlet ports may be smaller or larger than the number of cleaning devices. The inlet ports **224-I** are fluidly connected to corresponding outlet ports **166-I** on the operational tank **162**, that are fluidly connected to the discharge outlets (not shown) of the cleaning devices **160-I**. All the ports of the control manifold **210** may be camlocks. In one application, only ports **218** and **412** are camlocks and ports **418-I** and **224-I** are hammer unions. In one application, a hammer union is a connection that joins two parts with a threaded nut, and is commonly used in the oil and gas industry. The nut has protrusions that can be hit with a sledgehammer to tighten the connection and energize the seals.

[0035] In addition, the control manifold **210** includes the venturi eductor **450**, which is fluidly connected at a first inlet **451**, to the water supply pipe **414**, through the control valve **422**. The venturi eductor **450** has a second inlet **453** that is fluidly connected to a ball valve (i.e., a valve that has on and off positions) **454**. The second inlet **453** may also be fluidly connected to a crossover branch **460**, which may include one or more valves **462** (e.g., ball valve). The crossover branch **460**, when operational (i.e., the valve **462** is open), diverts the discharged slurry from the discharge block **430** to the eductor **450**. The ball valve **454** is connected to a suction port **456**, which is configured to be fluidly connected a hose **281**, which, in this embodiment, skims the water surface of the sediment storage tank **272** of the sediment storage truck **270**. The end of the hose **281** may be provided with a skimmer device **275**, which floats at the surface of the water **273** inside the tank **272**. The eductor **450** also has an outlet port **458**, which may be fluidly connected to another flexible hose **267** for supplying the water from the tank **272** to the water tank **262** of the water supply truck **260**. In other embodiments, the outlet port **458** of the eductor may be connected to a tank to discharge the sediment, as discussed later.

[0036] When in use (i.e., valve **422** is open), the eductor **450** is configured to receive, from the water supply pipe **414** of the water supply block **410**, a certain water flow. This water flow is pumped by the pump **264** of the water supply truck **260**, in addition to the water flow necessary for the cleaning devices **160-I**. The water flow at the first inlet **451** creates a negative pressure at the suction port **456** of the eductor (a vacuum), which draws in the separated water **273** from the sediment storage truck **272**. This water is then expelled by the eductor at the outlet **458**, and supplied to the water tank **262** of the water supply truck **260**. In this way, by diverting and using a fraction of the water flow from the water supply truck **260** to the eductor **450**, it is possible to recycle the water from the sediment storage truck **270**, which reduces the need for providing new water to the water supply truck **260** and for the sediment storage truck **270** to discharge its load. This method allows excess water that is carried in the slurry, from the cleaning devices **160**, to be returned to the water supply truck **260**, while the sediments from the slurry settle and remain in the

tank on the sediment storage truck **270**. This approach extends the capacity of both the water supply truck and sediment storage truck to enable multiple cleaning device to be operated until the sediment storage truck is filled with solids to a point that it reaches its maximum carrying weight of solids.

[0037] The above embodiments assume that the operational tank **162** is operating under a given pressure due to its role in the plant to which it belongs. This pressure helps to discharge the slurry from the tank, into the discharge block **430**. However, there are situations when the operational tank **162** is not under pressure, or its operation pressure is not enough to discharge the slurry into the discharge block **430**. For these situations, the crossover branch **460** of the control manifold **210** is used to fluidly connect the discharge pipe **212** to the suction port **453** of the eductor **450**. Thus, this feature enables operation of the cleaning devices **160-I** in a low pressure tank as the eductor **450** provides the necessary negative pressure for discharging the slurry from the operational tank **162**. This configuration is illustrated in FIG. 5. Note that for this configuration, the choke valve **222** is closed, so that there is no need for a hose to be connected to port **218**, for discharging the slurry to the sediment storage truck **270**. The slurry is now discharged through the crossover branch **460**, eductor **450**, and conduit **280** to sediment storage tank **272** of the sediment storage truck **270**. No water is recycled for this embodiment, from the sediment storage tank to the water supply tank as the eductor **450** is used to pump out the slurry.

[0038] In another embodiment, as illustrated in FIG. 6, a collection tank **600** is added to the system to replace the sediment storage truck **270**. Thus, although the sediment storage truck **270** is shown in the figure, its presence is not necessary. The sediment storage truck **270** may be used to empty the collection tank **600**, by using, for example, a pump and a conduit **280** connected to the bottom of the collection tank. The outlet port **218** of the discharge block **430** is fluidly connected through a pipe (or hose) **602** to the collection tank **600** for discharging the slurry, which separates into the sediment **271** and water **273**. A pipe (or hose) **604** may fluidly connect the inside of the collection tank **600**, at a height where it is expected to be only water, to the suction port **456** of the eductor **450**. In one variation of this embodiment, the skimmer device **275** may be used to maintain the end of the pipe **604** at the water surface inside the collection tank **600**. For this case, the end of the pipe **604** is flexible, so that it follows the skimmer device **275** as it moves with the water surface inside the tank.

[0039] The collection tank **600** in this embodiment provides two advantages to an oil and gas producer. The first advantage permits the system to be operated safely in the presence of high levels of potentially harmful gases, such as H₂S, methane, or others, that may be present in the slurry, which is removed with the cleaning devices **160-I**. The collection tank **600** also allows a connection **604**, from the top of the connection tank, back to an onsite flare or tank vapor recovery system **610**, that safely captures the harmful gases. The second advantage of the collection tank **600** is the elimination of the sediment storage truck **270** from the site, during operation of the cleaning devices. The solids removed with the cleaning devices can be stored onsite, in the collection tank **600**, and removed periodically, with the sediment storage truck. This may lower trucking costs for the operator of the operational tank **162**.

[0040] In another embodiment, as illustrated in FIG. 7, the water supply truck **260** is not present. For this embodiment, a pump **710** is provided next to a local water source **720** (e.g., associated with the plant or refinery that runs the operational tank **162**) and provides the necessary water to the control manifold **210**. In this embodiment, the water source **720** may be a tank or a water supply line. The pump **710** is fluidly connected to the inlet port **412** of the water supply block **410**. For this embodiment, the suction valve **456** of the eductor **450** is connected to the water of the sediment storage tank **272**, of the sediment storage truck **270**, similar to the configuration shown in FIG. 4, so that water from the sediment storage truck **270** is recycled to the water supply **720**, along a return water pipe **722**. Thus, in this embodiment, the pump **710** delivers water **273** to the cleaning devices **160-I** and also provides water to the eductor **450** to recover water from the sediment

storage tank **272** and return it to the onsite water supply **720**.

[0041] In yet another embodiment, as illustrated in FIG. **8**, it is possible to have neither the water supply truck **260** nor the sediment storage truck **270** as the local water supply **720** of the embodiment of FIG. **7** is used for supplying the water to the control manifold **210**, and the collecting tank **600** of the embodiment of FIG. **6** acts as the sediment storage tank. Although FIG. **8** shows the sediment storage truck **270**, the sediment storage truck **270** is not required for the operation of the cleaning devices **160-I**. The sediment storage truck **270** is shown in this figure to suggest that the collection tank **600** may be emptied, when desired, with the sediment storage truck **270**. In this embodiment, the eductor **450** is used to recycle the water from the tank **600**, i.e., to bring it back to the water source **720**.

[0042] One example of a cleaning device **160-I** used to clean the operational tank **162** discussed above is illustrated in FIG. **9**. Note that other cleaning devices may be used with the control manifold **210**. FIG. **9** presents a cleaning device **160-I** that is based on cyclonic jetting. The cleaning device **160-I** is configured to deliver a fluid (e.g., water) inside the tank **162** and extract a slurry from the tank. During delivery, the fluid may be a liquid, a gas, or a mixture, and during extraction, the fluid may be part of the slurry. In this embodiment, the cleaning device **160-I** includes a slurry intake assembly **905** coupled to an inlet line **906** and an outlet line **908**. During operation, inlet line **906** serves as a flow path for delivering the water (or other fluids) from the water supply truck **260**. In this and some other embodiments, inlet line **906** includes flexible piping. During operation, outlet line **908** serves as a flow path for the slurry discharge to the sediment storage truck **270**, which is retrieved through intake assembly **905**. In this and some other embodiments, outlet line **906** includes flexible piping. In some embodiments, inlet line **906** or outlet line **908** is formed with rigid piping lacking flexible piping.

[0043] When multiple cleaning devices **160-I** are used together in the operational tank **162**, the cleaning devices **160-I** may be flow balanced or internally flow balanced with respect to each other by using equal lengths of outlet lines **908** or by another arrangement. Some of those or other embodiments with multiple cleaning devices **160-I** are configured to have equal lengths of inlet lines **906**, making the inlet lines flow balanced. In some embodiments, inlet line **906** and outlet line **908** are considered to be members of slurry intake assembly **905**.

[0044] Slurry intake assembly **905** includes a hub manifold assembly **910** coupled to a plurality of fluidizers **924** by pairs of distribution lines **912** and gathering lines **922** for fluid communication. The hub manifold assembly **910** includes an inlet manifold **910A**, having a combined flow port **976** coupled to inlet line **906** as a manifold inlet port for fluid communication, and having multiple distributed ports **964**, with each distributed ports **964** coupled to one of the distribution lines **912** for fluid communication. The hub manifold assembly **910** further includes an outlet hub manifold **910B**, having a combined flow port **976** coupled to outlet line **908** as a manifold outlet port for fluid communication, and having multiple distributed ports **964**, with each distributed ports **964** coupled to one of the gathering lines **922** for fluid/slurry communication. Because distribution lines **912** and gathering lines **922** are flexible, they can be moved relative to hub manifolds **910A**, **910B** while they are fluidically connected to hub manifolds **910A**, **910B**. As a result, fluidizers **924** may be repositioned while remaining fluidically coupled to lines **912**, **922** and hub manifolds **910A**, **910B**. In some embodiments, the distribution lines **912** and gathering lines **922** are flexible hoses. In some embodiments, the slurry intake assembly **905** is flow balanced by using equal lengths of gathering line **922** for the multiple fluidizers. Some of these or other embodiments of slurry intake assembly **905** are configured to have equal lengths of distribution lines **912**, making the distribution lines flow balanced.

[0045] Continuing with FIG. **9**, each fluidizer **924** includes a fluid inlet **932** to receive a supplied fluid, includes an open end **926** to discharge the supplied fluid **980** into a volume and to generate a slurry **982**, by mixing the particulates in the tank with the fluid, and to extract/receive the slurry **982**, from the volume, and includes a slurry exit **944** to deliver the received slurry to a destination

(outlet hub manifold **910B**). The slurry received from the volume may be similar to the supplied fluid, or may be a solution with dissolved solids or a slurry formed from the supplied fluid, as examples. The open end **926** of fluidizer **924** includes an annular fluid exit and a slurry inlet, but in some embodiments, open end **926** and the fluid exit are configured differently than an annular passage disposed about the slurry inlet.

[0046] Fluidizer **924** includes a mounting feature, which in this example is a mounting bracket **946** disposed on the body of each fluidizer **924** and spaced apart from fluid inlet **932** and slurry exit **944**. For each fluidizer **924**, the fluid inlet **932** is coupled to a distribution line **912** to receive a supplied fluid from the inlet hub manifold **910A**, which is therefore arranged to perform as a fluid distribution source, and slurry exit **944** is coupled to a gathering line **922** to deliver the received slurry/fluid to the outlet hub manifold **910B**, which is therefore arranged to perform as a gathering manifold with respect to the multiple fluidizers **924**.

[0047] The cleaning device **160-I** shown in FIG. **9** includes four fluidizers **924**. Some other embodiments may have more or fewer fluidizers **924**, and the hub manifold assembly **910** (or equivalently **910A**, **910B**) may be modified or partially capped to accommodate the different quantity of fluidizers. As examples, in various embodiments, a hub manifold assembly **910** may be modified and coupled to two, three four, or five fluidizers, and some embodiments may accommodate six, seven, or eight fluidizers **924** or more or similar quantities of fluidizers of another design. Slurry intake assembly **905** may be flowed balanced for the several fluidizers. Other embodiments may include still more fluidizers coupled to hub manifold assembly **910**, up to a practical limit based on space or another engineering principle. As an example, the successful use of any particular number of fluidizers will depend, at least in part, on the pumping power available for delivering fluid to slurry intake assembly **905**.

[0048] It should be noted that hub manifold assembly **910** and slurry intake assembly **905** may be coupled to a quantity of fluidizers that is not divisible by the number two. Some embodiments have an even number of fluidizers while other embodiments have an odd number fluidizers coupled to an appropriately configured hub manifold assembly **910**, which may be configured and plumbed to achieve balanced flow among the multiple fluidizers. The balancing of flow may be achieved whether the hub manifold assembly **910** is distributing flow to or receiving fluid from the multiple fluidizers. Typically, another similar hub manifold assembly **910** with the same number of connections would be installed to perform the other task, for example, to receive or distribute flow.

[0049] Alternatively, some embodiments of slurry intake assembly **905** may instead include multiple hub manifolds plumbed to perform as outlet manifolds and a single, larger hub manifold with a sufficient number of distributed ports **964** could be installed to perform as an inlet manifold. Thus, for a particular slurry intake assembly, a larger, inlet hub manifold may have more distributed ports than any of the multiple outlet hub manifolds positioned to receive fluid or slurry from the fluidizers. The opposite arrangement is also possible, using multiple inlet hub manifolds in conjunction with a single, larger capacity outlet hub manifold. The flexibility in the selection of the number of flow paths and coupled fluidizers is based on the hub configuration of the hub manifold assembly **910**, which includes a plurality of circumferentially distributed ports **964** to achieve a circumferentially distributed flow pattern. Circumferentially distributed ports **964** may uniformly distribute fluid to or gather fluid from multiple lines simultaneously. Ports **964** may also be arranged to follow another pattern, different from the circumferential pattern. The incoming and outgoing flow paths depend on how a particular hub manifold is plumbed to adjacent equipment. In the embodiment shown, hub manifolds **910A** and **910B** are further augmented by the use of flexible lines between the hub manifolds and the fluidizers. The multiple, circumferentially distributed ports **964** configure the hub manifold assembly **910** to have a generally radial pattern of flow paths leading to or emanating from a shared outlet or inlet flow path.

[0050] In FIG. **9**, distributed ports **964** of hub manifolds **910A**, **910B** are coupled to the flexible distribution lines **912** and flexible gathering lines **922**, respectively. By these connection lines,

fluidizers **924** are each coupled to a distributed port **964** of distributing hub manifold **910A** and to a distributed port **964** of gathering hub manifold **910B** for fluid communication. In FIG. **9**, each distributed port **964** is coupled to a single fluidizer **924**.

[0051] The independent control manifold **210** discussed in the previous embodiments has been implemented on the platform **230**, thus forming a control manifold assembly **200**. This is advantageous because the control manifold **210** becomes independent of the trucks **260**, **270** or the operational tank **162**, which means that the control manifold **210** may be moved at any desired operational tank, independent of the presence of the trucks (i.e., it is portable). This not only reduces the cost of removing the slurry from the tanks, but also reduces a time necessary to prepare the entire system for sediment removal and also the capital investment for the operator of the control manifold.

[0052] However, in another embodiment, as illustrated in FIG. **10**, it is possible to place the control manifold **210** on one of the trucks **260** or **270**. The control manifold **210** may be directly attached to a chassis **1010** of the truck or to the tank **262/272**. In one embodiment, a control manifold **1110** may be implemented, as illustrated in FIG. **11**, on the truck **260/270**, with a similar or different configuration than the control manifold **210**. The control manifold **1110** includes some of the elements of control manifold **210**, as illustrated in FIG. **12**, but differently connected. In addition, the control manifold **1110** includes a bypass valve **1210** that may fluidly connect the inlet port **412** (for the water supply) to the inlet port **224** (for receiving the slurry from the cleaning devices **160-I**). Note that the inlet port **224** in this embodiment is configured to act as a valve, e.g., ball valve. While FIG. **12** shows a single port **224**, in one embodiment, it is possible to have plural ports **224-I**, as shown in the embodiment of FIG. **4**, depending on the number of cleaning devices to be serviced.

[0053] In operation, the various valves of the control manifold **1110** are controlled as follows. If the control manifold **1110** is used for sediment removal from an operational tank under pressure, then the water supply from the water supply truck is provided to port **412**, the valve **422** that controls the water flow to the eductor **450** is closed, and the valve **416** that controls the water flow to the intake of the cleaning device **160-I** is open, so that the water is supplied to the cleaning device. The bypass valve **1210** is closed so that the supplied water does not mix up with the slurry received at port **224**. The port/valve **224** is open for this scenario so that the slurry flows along pipe **212** to connection port **218**, which is fluidly connected to the sediment deposit tank **272** of the truck **270** or the collection tank **600**.

[0054] However, if the operational tank **162** is not under pressure, or its pressure is too low to pump the slurry to the port **224**, then the control manifold **1110** is reconfigured to (partially) close valve **416**, (partially) open valve **422**, so that a water flow is provided to the eductor **450**, to provide a vacuum effect in the pipe **212**, to suck the slurry from the port **224**, through the valve **462**. Port **218** is closed for this scenario. In this case, the sediment storage tank **272** of truck **270** is connected to outlet port **458**, and the outlet port **458** is opened, so that the slurry from the operational tank **162** flows to the sediment storage tank **272**.

[0055] The valves shown in FIG. **12** may be further configured to allow a backflush of the cleaning device **160-I**. For example, if the operational tank is under pressure, the bypass valve **1210** is opened, valves **422** and **416** are closed, port **218** is closed, and valve **462** is closed, and thus, the water supplied at port **412** moves through port **224** to the slurry line of the cleaning device for backflushing. If a flushing of the control manifold to the sediment storage tank **272** is desired, the bypass valve **1210** is opened, valves **416** and **422** are closed, port **224** is closed, valve **462** is closed, and port **218** is open, so that the flush liquid moves along pipe **212** and exits port **218** and then moves into the sediment storage tank **272** of the sediment storage truck **270**.

[0056] For the backflush of the operational tank when under low pressure, the water is provided at port **412**, valve **416** is closed, valve **422** is opened, port **224** is open, valve **462** is partially opened, port **458** is open and connected to the sediment storage tank, and port **218** is closed. For the

backflush of the slurry line of the cleaning device **160-I** when the operational tank **162** is at low pressure, all the ports and valves are closed except for bypass valve **1210** and port **224**.

[0057] FIG. **13** illustrates a system in which a single truck **260** or **270** is provided with the control manifold **1110**, which is attached outside the water tank **262**. The figure also shows the connections to the operational tank **162**, and the connections between these various elements. More specifically, FIG. **13** shows the skimmer device **275** floating in the water **273** inside the tank **262/272** of the truck **260/270**. Note that initially, the water tank **262** for this embodiment is full of water, and, as the cleaning process of the operational tank **162** is progressing, sediment **271** starts to accumulate in the water tank **262**, as a single truck **260** or **270** and a single tank **262** or **272**, are used. The pump **264** is fluidly connected to the skimmer device **275**, for pumping the clean water **273**, along pipe/hose **266**, to the intake port **164** of the functional tank **162**. A motive valve **1310** may be provided to the inlet line **906** for each cleaning device **160-1**, **160-2** (only two shown in the figure, but more or less may be used for a given operational tank) in the operational tank **162**. The outlet line **908** of each cleaning device is provided with a corresponding slurry isolation valve **1312** and a slurry operating valve **1314**, as illustrated in FIG. **13**. In one embodiment, these valves are optional. A bypass valve **1316** may fluidly connect the inlet port **164** to the connection port **166**, as also shown in the figure. In this embodiment, the sediment extracted from the functional tank **162**, along hose **280**, is discharged inside the tank **272** at the connection port **218**. Note that the fluid connection between the connection port **218** and the interior of the tank **272** may be achieved by a pipe connection **1320**, as schematically illustrated in FIG. **13**.

[0058] A method **1400** for removing the sediment **271**, with the control manifold **210**, from the operational tank **162**, is now discussed with regard to FIG. **14**. The method **1400** includes a step **1402** of supplying a first cleaning fluid (e.g., water, or a mixture of water and surface tension reducer, or other liquids), with the supply block **410** of the control manifold **210**, to an inlet **164** of the operational tank **162**, a step **1404** of discharging the sediment **271**, with the discharge block **430** of the control manifold **210**, from an outlet **166** of the operational tank **162**, a step **1406** of diverting a part of the first cleaning fluid **273**, from the supply block **410** to a first inlet **451** of a venturi eductor **450** to create a vacuum at a second inlet **453**, and a step **1408** of pulling in a fluid, at the second inlet **453** of the venturi eductor **450**.

[0059] In one embodiment, the fluid is a second cleaning fluid (e.g., water) from an external tank, which is filled with water and sediment, as illustrated in FIGS. **4**, **6**, **7**, and **8**. In another embodiment, the fluid is a mixture of water and sediment extracted from the operational tank, as illustrated in FIG. **5**.

[0060] The method may further include a step of providing an output of the venturi eductor to a water only supply tank, as shown in FIGS. **4**, **6**, **7**, and **8**. Alternatively, the method may include providing an output of the venturi eductor to a water and sediment tank, as shown in FIG. **5**.

[0061] The method may also include a step of recycling the first cleaning fluid, from a tank that stores the sediment, by skimming the first cleaning fluid from the tank and supplying the skimmed first cleaning fluid to the supply block. In this or another embodiment, the sediment is discharged in a same tank from which the first cleaning fluid is supplied.

[0062] While a number of embodiments have been shown and described, modifications thereof can be made by one of ordinary skill in the art without departing from the scope or teachings herein. The embodiments described herein are examples only and are not limiting. Many variations, combinations, and modifications of the systems, apparatuses, and processes described herein are possible and are within the scope of the disclosure. Accordingly, the scope of protection is not limited to the embodiments described herein, but is only limited by the claims that follow, the scope of which shall include all equivalents of the subject matter of the claims. The inclusion of any particular method step or operation within the written description or a figure does not necessarily mean that the particular step or operation is necessary to the method. The steps or operations of a method listed in the specification or the claims may be performed in any feasible

order, except for those particular steps or operations, if any, for which a sequence is expressly stated. In some implementations two or more of the method steps or operations may be performed in parallel, rather than serially.

[0063] One of ordinary skill in the art will understand that the above description has broad application, and the discussion of any embodiment is meant to be exemplary of that embodiment, and is not intended to suggest in any way that the scope of the disclosure, including the claims, is limited to that embodiment.

[0064] The figures are not drawn to scale. Certain features or components disclosed herein may be shown exaggerated in scale or in somewhat schematic form, and some details of some elements may not be shown in the interest of clarity and conciseness. In some of the figures, in order to improve clarity and conciseness, one or more components or aspects of a component may be omitted or may not have reference numerals identifying the features or components. In addition, within the specification, including the drawings, like or identical reference numerals may be used to identify common or similar elements.

[0065] As used herein, including in the claims, the terms “including” and “comprising,” as well as derivations of these, are used in an open-ended fashion, and thus are to be interpreted to mean “including, but not limited to . . .” Also, the term “couple” or “couples” means either an indirect or direct connection. Thus, if a first component couples or is coupled to a second component, the connection between the components may be through a direct engagement of the two components, or through an indirect connection that is accomplished via other intermediate components, devices and/or connections. The recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be based on Y and on any number of other factors. The word “or” is used in an inclusive manner. For example, “A or B” means any of the following: “A” alone, “B” alone, or both “A” and “B.” In addition, the word “substantially” means within a range of plus or minus 10%.

[0066] Furthermore, any reference to a relative direction or relative position is made for purpose of clarity, with examples including “top,” “bottom,” “up,” “upper,” “upward,” “down,” “lower,” “clockwise,” “left,” “leftward,” “right,” “rightward,” “internal,” and “external”. For example, a relative direction or a relative position of an object or feature may pertain to the orientation as shown in a figure or as described. If the object or feature were viewed from another orientation or were implemented in another orientation or position, it may then be helpful to describe the direction or position using an alternate term.

[0067] As used herein, “piping” and “line,” when referring to plumbing, may be used interchangeably and may include any of the following, whether alone or in any combination: pipe, tubing, hose, fittings, or any other tubular member suitable for containing or moving a fluid in a system described herein. Therefore, the “piping” or “line” may include, as examples, any valve(s), instrument(s), or port(s) for controlling, measuring, or accessing a fluid in the line. The term fluid circuit may be descriptive of piping or a line disclosed herein. Other potential variations and supplements will be understood by those skilled in the art.

Claims

1. An independent control manifold for handling supplying a cleaning fluid to, and extracting a sediment from an operational tank, the control manifold comprising: a supply block configured to receive the cleaning fluid from an external first tank and to supply the cleaning fluid to the operational tank; a discharge block configured to receive the sediment from the operational tank and to supply the sediment to an external second tank; and a venturi eductor that fluidly connects the supply block to the discharge block, and is configured to either extract the sediment from the operational tank or to recover the cleaning fluid.
2. The control manifold of claim 1, wherein the supply block comprises: an inlet port configured to

connect to a first hose for receiving the cleaning fluid; a control valve fluidly connected to the inlet port; one or more outlet ports configured to connect to an inlet of a cleaning device; and piping fluidly connecting the inlet port to the control valve and the control valve to the one or more outlet ports.

3. The control manifold of claim 2, wherein the discharge block comprises: one or more inlet ports configured to connect to an outlet of the cleaning device; a choke valve fluidly connected to the one or more inlet ports; a connection port configured to connect to a second hose for discarding the sediment; and piping fluidly connecting the one or more inlet ports to the choke valve and the choke valve to the connection port.

4. The control manifold of claim 3, further comprising: a cross-over branch that fluidly connects the piping of the discharge block to the venturi eductor, wherein the cross-over branch includes a valve.

5. The control manifold of claim 4, wherein the venturi eductor comprises: a first inlet configured to be fluidly connected to the supply block; a second inlet configured to be fluidly connected to the cross-over branch; and an outlet port configured to be fluidly connected to an external third tank.

6. The control manifold of claim 5, wherein the external third tank is the external first tank and the second inlet port is fluidly connected to the external second tank.

7. The control manifold of claim 5, wherein the external third tank is the external second tank and the second inlet port is closed.

8. The control manifold of claim 5, further comprising: a collection tank fluidly connected to the second inlet port of the venturi eductor, the collecting tank being the external second tank, wherein the first inlet port of the venturi eductor is fluidly connected to the external first tank, and wherein the connection port of the discharge block is fluidly connected to the collection tank.

9. The control manifold of claim 5, further comprising: a skimming device configured to skim the cleaning fluid from the external second tank and provide the skimmed cleaning fluid to the second inlet port of the venturi eductor, wherein the external second tank is the first external first tank.

10. The control manifold of claim 5, wherein the second inlet port of the venturi eductor is fluidly connected to the external second tank, the external second tank is the first external tank, and the connection port of the discharge block is fluidly connected to the external second tank.

11. The control manifold of claim 1, wherein the external second tank is one of a sediment storage tank associated with a sediment storage truck, the external first tank associated with a water storage truck, or a collection tank, which is independent of a truck.

12. The control manifold of claim 1, further comprising: a platform configured to support the supply block, discharge block, and venturi eductor so that the control manifold is mobile and autonomous.

13. A cleaning truck for supplying a cleaning fluid and extracting a sediment from an operational tank, the cleaning truck comprising: a tank for storing the cleaning fluid and the sediment; and a control manifold located outside the tank and comprising: a supply block configured to receive the cleaning fluid from the tank and to supply the cleaning fluid to the operational tank; a discharge block configured to receive the sediment from the operational tank and to supply the sediment to the tank; and a venturi eductor fluidly connecting the supply block to the discharge block, and configured to either extract the sediment from the operational tank or to recover the cleaning fluid from the tank.

14. A method for removing a sediment, with a control manifold, from an operational tank, the method comprising: supplying a first cleaning fluid, with a supply block of the control manifold, to an inlet of the operational tank; discharging the sediment, with a discharge block of the control manifold, from an outlet of the operational tank; diverting a part of the first cleaning fluid, from the supply block to a first inlet of a venturi eductor to create a vacuum at a second inlet; and pulling in a fluid, at the second inlet of the venturi eductor.

15. The method of claim 14, wherein the fluid is a second cleaning fluid from an external tank,

which is filled with water and sediment.

16. The method of claim 14, wherein the fluid is a mixture of water and sediment extracted from the operational tank.

17. The method of claim 14, further comprising: providing an output of the venturi eductor to a water only supply tank.

18. The method of claim 14, further comprising: providing an output of the venturi eductor to a water and sediment tank.

19. The method of claim 14, further comprising: recycling the first cleaning fluid, from a tank that stores the sediment, by skimming the first cleaning fluid from the tank and supplying the skimmed first cleaning fluid to the supply block.

20. The method of claim 14, wherein the sediment is discharged in a same tank from which the first cleaning fluid is supplied.
